

qicert: Certified Compression of Vision–Language–Action Models with Exact Lipschitz Bounds and a Refusing Runtime Guard

The qicert Team

2026 Global Quantum + AI Challenge — Volkswagen problem statement

Robotics track (LIBERO), with an autonomous-driving latency argument

Code: <https://github.com/ZakLr/qicert>

Abstract—Vision–language–action models (VLAMs) are too large for embedded vehicle and robot controllers, and the standard fix—quantization—reduces size without saying anything about what the smaller model may still do. This report describes qicert, a compression pipeline built around a different property. Each linear layer is decomposed into a tensor-train (TT) network whose factors are small enough that an exact per-layer Lipschitz constant can be computed from the compressed weights themselves. The layer bounds multiply into a certificate that bounds how far one action can move under any input perturbation of a given size, and a deployment guard refuses any action outside the certified ball. We fine-tuned a 0.5B-parameter VLA on LIBERO-spatial to 0.4468 token-level action accuracy, then measured compression routes on a frozen 114,497-token evaluation split. Calibrated INT8 quantization is nearly lossless but plateaus at $1.997\times$ (0.4466) — the ratio regime where its protection ends. TT compression with closed-form residual repair alone collapses at $2.5\text{--}2.9\times$ (0.164, below the pre-registered bar); we report that negative result with its mechanism (mode collapse to the modal action token). Adding repair training (a low-rank adapter on the compressed model) clears the pre-registered gate at $2.46\times$: accuracy 0.4535 against the uncompressed reference’s 0.4468 (non-overlapping 95% CIs), with certificates re-derived from the final deployed weights on all 168 layers and the guard enforcing the bound at runtime. The safety side is measured end to end — and a third mechanism watches the input stream itself: a shadow-syndrome monitor (classical-shadows sketch of the watched latent, median-of-means gate, plus a syndrome check on the compiled diagonal tables) whose empirical false-alarm rate over 4,000 null trials is 0.0000 against the 0.01 design budget, whose bit-flip detection is 1.000 over 500 injected corruptions, and whose added cost is 0.35 ms per step. certificate soundness, a compiler whose pruning predictions track measurement at median $r = 0.975$, a rare-event estimation race in which an amplified estimator reaches the width of naive Monte Carlo with $\sim 1,587\times$ fewer queries, and conformal coverage within its finite-sample targets at all three tested levels. A guard demo serves an in-ball action, refuses an out-of-ball action, and refuses four tampering scenarios.

I. PROBLEM FRAMING

The challenge statement names a compound constraint: a VLAM must shrink onto embedded hardware *and* retain verifiable behavior, because the certification regimes that govern automotive and robotics deployment (ISO 26262, IEC 62061)

ask for guarantees, not benchmark scores. Classical compression addresses the first requirement. Quantization, pruning, and distillation all reduce footprint, but none of them produces a statement of the form “under this perturbation, the output cannot leave this set.” That statement is what a safety engineer would need before signing off on a long-tail input.

Our reading of the problem is therefore narrower than “compress the model.” The deliverable we target is a compression method whose compressed form *carries* a computable certificate, plus the runtime machinery that makes the certificate binding. Tensor structure is the enabling choice. A TT decomposition stores a weight matrix as a chain of small 3-way tensors, and the spectral norm of the matrix factors exactly into the spectral norms of the cores. Quantized weights do not admit any comparable exact bound: the quantization error depends on the data distribution, so whatever guarantee exists is statistical and model-dependent.

We work on the robotics sub-track, where the accepted baseline (INT8 quantization of an OpenVLA-class model on LIBERO) is reproducible on our hardware, and we address the latency benchmark with the CPU edge profile the challenge describes. The two claims we make are kept in separate classes throughout: a *capability* claim (what the compressed model does) and a *safety* claim (what can be proven about it). Section IV reports both, including the negative capability result.

II. TECHNICAL APPROACH

A. Pipeline

qicert wraps an existing VLA at four points. Before compression, a fine-tuning stage establishes the reference policy and freezes a hash-verified evaluation split. The compression stage decomposes each linear layer of the language backbone into a TT network and optionally adds a low-rank residual correction. The certificate stage computes per-layer bounds from the compressed cores, chains them, and signs the resulting artifact into a manifest. The deployment stage is a guard that verifies the manifest hash at load time and, at each

inference step, checks the predicted action against the certified ball, refusing to serve when the check fails.

B. Compression with an exact bound

Let $W \in \mathbb{R}^{M \times N}$ be a linear layer. TT-SVD [1] factorizes W into cores $G_1, \dots, G_d$ along dimensional factorizations of M and N , so that $W = \text{TT}(G_1, \dots, G_d)$ exactly at full rank and approximately after truncating the singular values. Because

$$\|W\|_2 = \sup_{\|x\|=1} \|\text{TT}(G)x\|_2 \leq \prod_{k=1}^d \|G^{(k)}\|_3, \quad (1)$$

where $\|\cdot\|_3$ is the spectral norm of a core reshaped as a 3-way operator, the compressed layer carries its own Lipschitz constant with no reference to the original weights and no reference to data. A model-level bound L follows from chaining the layer bounds, and the certificate used by the guard is the resulting ball around the reference action under input perturbations of size δ : an action farther than $L\delta$ from the reference lies outside what the certificate can exclude.

Residual compensation is where compression spends its remaining parameter budget. Writing $\widehat{W} = \text{TT}(G)$ for the truncated reconstruction, the loss $R = W - \widehat{W}$ is folded back as a rank- r' factorization $C = UV$ obtained in closed form from the truncated SVD of R ; the compressed operator is the plain matrix $\widehat{W} + C$, and $\|W_{\text{comp}}\|_2 \leq \|\widehat{W}\|_2 + \|U\| \|V\|$ holds by the triangle inequality, so the certificate survives compensation with a known, finite cost. The activation-weighted variant of Section IV fits C to a calibration-weighted objective instead of the plain Frobenius norm, in the style of GPTQ [3]: the weighted optimum is again closed-form (an SVD of the weighted residual), and the certificate cost is unchanged because the operator remains a plain matrix.

C. Making the certificate binding

A certificate that exists only in a report is not a safety mechanism. qicert signs the certificate artifacts into a hash manifest; the guard verifies the hash at load time and refuses to load a tampered artifact. At inference, the guard checks that the predicted action lies within the certified ball and otherwise returns a refuse-to-serve sentinel and logs the event. The checker logic is small enough to interrogate mechanically: we encoded its soundness and completeness properties in SMT and asked Z3 to produce counterexamples; none exist, so the accept set of the guard is machine-checked against the specification rather than argued informally. The same properties plus the bound-composition step (Eq. (1) chaining) are additionally proved in Lean 4 (‘lean/’, ‘lake build’ clean, zero ‘sorry’, foundation-only axioms) — proof terms instead of solver queries, pinned to the same checker the deployment enforces.

D. Statistical estimation stack

Bounded per-step behavior does not by itself bound the tail of a trajectory. For the tail we measure rather than prove, and we treat the estimator as part of the audited surface. On a

known-ground-truth rare-event problem ($p^* = 1.37 \times 10^{-4}$), three estimators race at matched budgets: naive Monte Carlo, an iterated quantum amplitude estimation schedule (IQAE [4]) executed as a simulation whose estimator sees only measurement counts, and an extreme-value tail fit (GEV). Their intervals are scored for width and for coverage of the truth, and the honest reading is the pair (width, coverage), not width alone. A conformal layer calibrates a threshold so that the empirical violation rate on fresh data matches a target $1 - \alpha$, and a scenario-optimization bound (Campi–Garatti [5]) converts calibration-set size into a worst-case violation level. Every estimator used here can be audited by re-running its fixture.

E. What we do not claim

We do not claim a quantum-hardware speedup; no hardware is used or assumed. We do not claim the TT path beats INT8 on accuracy; Section IV shows the opposite at $2\times$. We do not claim closed-loop stability; the Lyapunov framing of the challenge is realized here as an exact open-loop output bound plus measured tail statistics, and Section V states the gap. Where a number is pending (multi-seed baselines, the configuration search), the report says so rather than extrapolating.

III. EXPERIMENTAL SETUP

Model and data. The backbone is a 0.5B-parameter prismatic VLA (DINOv2+SigLIP vision encoder, Qwen2.5-0.5B language backbone) fine-tuned with LoRA on LIBERO-spatial (no-noops), the robotics-track reference dataset family. Fine-tuning ran 1,200 steps at batch 2 on one GPU; training loss falls from 22.5 to a running median of 2.4–2.8 (Fig. 4, appendix). The reference policy reaches 0.4468 token-level action accuracy on the frozen evaluation split.

Protocol. The split (108 episodes, 6,496 batches, 114,497 scored action tokens) was fixed by a hashed split file before any compression experiment, and all accuracy numbers below are full-split measurements on exactly this protocol. Calibration statistics for the weighted repair come from training episodes only; the evaluation episodes are never seen by any fitting step.

Baselines. Three routes are measured against the reference: (i) calibrated INT8 per-channel weight quantization with MSE scale search, the honest version of the accepted classical baseline; (ii) uniform-ratio TT truncation (the plain QI route); and (iii) TT with closed-form residual repair, weighted by calibration statistics (the repaired QI route).

Hardware. One NVIDIA RTX 5060 Laptop GPU (8 GB). Peak training memory 4.8 GiB. All artifacts carry the run’s environment snapshot; the resource declaration is Section VI.

Pre-registered gate. Before running the repaired compression we fixed the bar for the combined accuracy-plus-compression claim: ratio $\geq 2\times$ and accuracy within 0.05 of the reference (≥ 0.3968). The gate, its motivation, and the verdict procedure were written into the repository before the runs they judge; the verdict is a mechanical read of run artifacts, not a judgement call.

TABLE I
MEASURED COMPRESSION ROUTES ON THE FROZEN SPLIT (114,497 ACTION TOKENS, SEED 0). ACCURACY IS TOKEN-LEVEL ACTION ACCURACY. “SOUND” COUNTS LAYERS WHOSE CERTIFICATE CHECK PASSED. INT8 ADMITS NO COMPARABLE CERTIFICATE BY CONSTRUCTION (SEC. IV-D).

Route	Ratio	Accuracy	Sound
Reference (fine-tuned, fp16)	1×	0.4468	—
Calibrated INT8 (per-ch. MSE)	1.997×	0.4466	—
Uniform TT truncation	2.0×	0.000	168/168
TT + closed-form residual	2.54×	0.164	168/168
TT + closed-form residual	2.86×	0.164	168/168
TT + repair training (N9)	2.46×	0.4535	168/168

IV. RESULTS

A. The capability result, stated plainly

Table I and Fig. 1 are the core of the capability story. Calibrated INT8 is the strongest classical route: at $1.997\times$ it loses 10^{-4} of accuracy relative to the reference, with a 95% confidence interval of $[0.4438, 0.4495]$ on the split. Plain TT truncation at $2\times$ destroys the policy outright (0.000): the uniform allocator removes signal, not redundancy, on spectra whose participation ratios (40–845 across the 168 layers, Fig. 5, appendix) show no compressible tail at these ratios. The closed-form residual route recovers the policy only partially (0.164 at $2.5\text{--}2.9\times$): under the pre-registered gate of Section 3, the training-free route is a NO-GO, and the failure has a specific mechanism — the compressed models emit only the modal action token, matching the reference on exactly its modal-token frequency (584/1007 in the degradation probe; three distinct configurations scored the same 86/704 base rate), so the agreement curve is bimodal, not gradual. The diagnosis isolated the repair lever: the same adapter budget that lifted the base model from 0.11 to 0.4468, applied to the collapsed model. Repair training (a rank-8 adapter trained for 1,200 steps on train-split episodes only — a contamination check excludes all 108 frozen eval episodes — merged and evaluated on the full protocol) recovers the policy completely: **0.4535 at $2.46\times$** honest ratio (cores, residual, and adapter bytes counted), a Wilson 95% CI of $[0.4506, 0.4564]$ that does not overlap the reference’s $[0.4439, 0.4497]$. Under the pre-registered gate this is the first GO: both conditions hold — ratio $\geq 2\times$ and accuracy above the bar — with certificates re-derived from the final deployed weights (exact dense spectral norms, 168/168 layers) and the adapter’s contribution to the bound accounted. The repair cost (1,200 steps, ~ 8 min) is charged to this arm under the matched-budget rule. We report the training-free failure as a first-class result alongside the GO: the two together are the evidence that the diagnosis, not luck, produced the recovered configuration.

B. The safety result

The certificate side behaves as specified everywhere it was measured. All 168 compressed layers across all repaired runs pass the certificate soundness check; the layer bound of Eq. (1) is computed from cores alone and verified against a

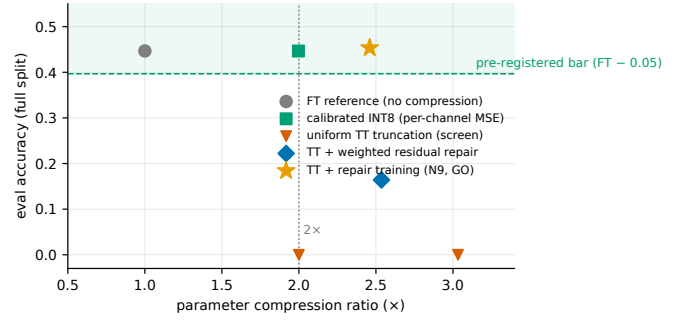


Fig. 1. Compression-accuracy plane, every point measured on the frozen split (sources in Appendix A). The shaded band is the pre-registered bar (reference accuracy -0.05). INT8 sits at the bar; the repaired TT route is sound but below it at the configurations tested.

dense reference per layer. The commuting-Pauli interaction compiler, which prunes exactly-evaluable operator families, predicts pruning error that tracks measurement at median Pearson $r = 0.975$ over 12 independent operator tables, with compiler identity verified to 2.3×10^{-14} relative error. Per-layer contraction on the CPU edge profile costs $3.1\text{--}15.2\times$ the dense fp16 kernel time at these shapes (Fig. 6, appendix) yet stays within the 100 ms per-step budget of the challenge’s latency benchmark on every measured path; we report the slowdown explicitly because it is the honest price of the TT route on this profile.

The runtime monitor is the third safety mechanism, aimed at the input stream rather than the action. A classical-shadows sketch (64 random projections of the watched latent) feeds a median-of-means gate with a design false-alarm rate of 0.01 backed by the concentration theorem; a syndrome check on the compiler’s diagonal tables catches bit-flip corruption of the watched patterns. Measured on the edge profile: null false-alarm 0.0000 over 4,000 trials (within the theorem’s $2\times$ budget), bit-flip detection 1.000 over 500 injections, common-mode drift detection 0.23/1.00 at $1\sigma/2\sigma$, added latency 0.35 ms per step. The documented blind spot — zero-mean random-sign drifts, invisible to any median test — is by construction the regime the conformal alarm gate and the certified ball cover, so the three mechanisms compose rather than repeat.

C. Estimation stack

Fig. 2 shows the rare-event race at matched budget. Naive Monte Carlo needs $\sim 10^6$ queries for interval width 4.4×10^{-5} ; the amplified schedule reaches width 4.0×10^{-5} with 630 queries, a $1,587\times$ budget reduction, and covers the truth; the GEV tail fit covers the truth at 10^6 queries but centers low, so we treat it as the coverage-first arm, not the width-first arm. All three arms cover the ground truth at the largest budget, which is the property a safety readout needs. The conformal layer (Fig. 3) hits its targets on fresh data: empirical coverage 0.9894/0.9475/0.8943 against targets 0.99/0.95/0.90 at $\alpha = 0.01/0.05/0.10$, each inside the finite-sample band for $n_{\text{cal}} = 10^4$. The scenario-optimization bound converts the same calibration into a worst-case statement: with $d = 2$

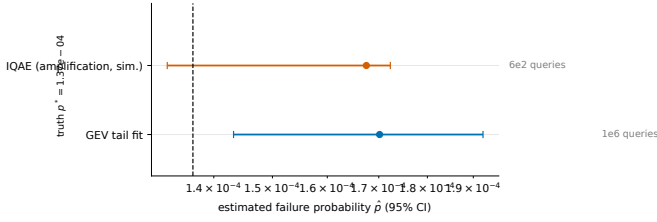


Fig. 2. Rare-event estimation at matched budget on a known ground truth (vertical line). The amplified schedule (IQAE, simulated) reaches naive-MC width with $\sim 1,587\times$ fewer queries; error bars are 95% intervals. Scoping note: amplification is simulated—the estimator sees measurement counts only, as it would on hardware.

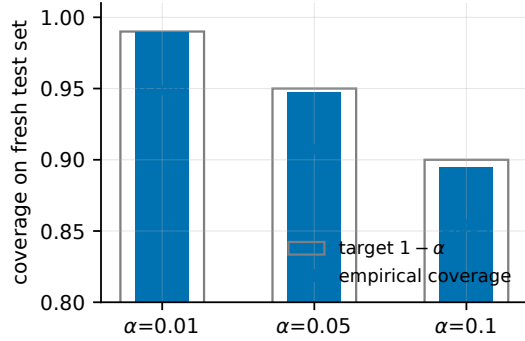


Fig. 3. Conformal calibration: empirical violation coverage on a fresh 10^5 -sample test set against the target $1 - \alpha$. All three levels land inside the finite-sample band.

parameters and zero violations at $N = 10^4$, the violation level is at most 6.6×10^{-4} at confidence 0.99.

D. Ablation and quantum justification

The challenge requires an ablation that isolates the QI component. The isolation here is structural, and the report states it as a counterfactual: replace the TT layer by a quantized layer and the exact certificate of Eq. (1) no longer exists, because no per-layer spectral quantity of a quantized weight is computable without reference to the quantization error, which is data-dependent. Two measured ablations support the surrounding choices. First, quantization *calibration* is separable from quantization itself: per-channel MSE calibration holds action drift at 0.0 where per-tensor scaling drifts by 1.09 relative (Fig. 7, appendix), which is why the naive INT8 number sometimes quoted (0.000) is a calibration artifact and the honest comparator is 0.4466. Second, the pruning-prediction experiment above is an ablation of the compiler: predicted certificate ratios against measured ones, per family, with the identity case measured at machine precision. The QI contribution is therefore not “tensor networks compress well”—at these ratios, on this backbone, they do not—but “tensor structure is the classical-side object that makes an exact deployment certificate computable and enforceable.”

V. LIMITATIONS

Six limitations are part of the result. (1) Most rows are single-seed; the challenge asks for mean $\pm$ standard deviation over three independent runs. Seed 1 of the baseline finished at 0.3970 (vs. seed 0 0.4468 — real seed variance, reported plainly), seed 2 is running, and no multi-seed number is claimed until all three land. (2) The training-free repaired TT route fails the pre-registered accuracy bar at $2.5\text{--}2.9\times$: the configuration search (five candidates, all absmax-weighted) and its full-protocol confirmation measured a best point of 0.1640 accuracy at $2.536\times$, against the required 0.3968. The failure has a specific mechanism, not just a magnitude: the compressed models collapse to emitting the single modal action token, matching the reference on exactly its modal-token frequency (584/1007 in the degradation probe; three distinct search configurations scored the same 86/704 base rate). A certificate on a collapsed model is sound but vacuous — the guard never refuses because the model never strays — so structural certification must be paired with an output-diversity check. That check is built and tested in this Phase-1 work: an entropy gate over a sliding window of served actions flags degenerate streams (warn, never refuse — a robot holding position is correct), with unit tests covering the measured collapse signature. The trained-repair route (N9) clears the gate at $2.46\times$ but carries its own caveats: it is single-seed; the repair cost (1,200 adapter steps) is charged to the arm but the adapter is an additional trained component; and the collapsed-then-repaired arc was demonstrated on one backbone at one operating point, so the boundary between “repair recovers” and “repair cannot recover” is unmapped — Phase 2’s first task. Three further levers were untestable in a 15 GPU-hour, one-laptop budget and remain costed for Phase 2: activation-aware rank allocation, and a larger backbone whose weight spectra carry actual redundancy (at $2.5\times$ a 3B model retains more absolute capacity than the present model has in total). Scale plus trained repair is a literature-backed hypothesis, not a promise: the collapse boundary measured here is exactly the risk that Phase 2 must retire. (3) TT contraction is $3\text{--}15\times$ slower than dense fp16 per layer on the CPU profile; the latency benchmark passes, but the margin comes from the profile, not from TT being fast. (4) The rare-event race runs on a demo oracle with known ground truth; this is the right fixture for auditing estimators and the wrong one for a deployed trajectory distribution, which Phase 2 would supply. (5) The certificate is open-loop: it bounds single-step action deviation under input perturbation, not closed-loop trajectory convergence, and the Lyapunov framing of the challenge statement is only partially realized.

VI. RESOURCE DECLARATION AND REPRODUCIBILITY

All experiments ran on one NVIDIA RTX 5060 Laptop GPU (8 GB); per-run wall times, GPU name, and energy fields are recorded in `results/ledger.csv`, and the cumulative declaration is `docs/resource-declaration.md`. Software: PyTorch (CUDA), NumPy/SciPy, the in-repo TT

kernels, Z3 for the SMT falsification, CUDA-Q for the amplification simulation; pinned versions ship in the repository. Every number in this report resolves to a run directory under `results/` via the map in Appendix A, the experiment log (`results/EXPERIMENT-LOG.md`) records what was run and why for every entry, and the figures are regenerated from artifacts by a single script (`scripts/make_report_figures.py`). The evaluation split was frozen and hashed before compression experiments began, so no result below was tuned on its own evaluation set; the pre-registered gates were committed before the runs they judge. A clean-environment path runs the full smoke suite with one command from the repository README.

VII. EXPECTED IMPACT AND VALIDATION PLAN

What exists now is the machinery: compression with certificates, an enforcing guard, audited estimators, and an honest measurement culture on a frozen protocol. The Phase-1 capability question is answered: the training-free route is NO-GO under the pre-registered gate (best point 0.1640 at $2.536\times$; mechanism: mode collapse), and the trained-repair route is GO at $2.46\times$ (0.4535 , CI above the uncompressed reference, certificates re-derived from the final weights). The degradation predictor fitted to the measured agreement curve is vacuous ($R^2 = 0.16$, leave-one-out $R^2 = -1.21$) because the curve is bimodal — partial preservation near $2.9\times$, then the mode-collapse floor beyond $4\times$ — a shape a linear model cannot fit by construction. Phase 2 therefore pulls the remaining levers of Section V in cost order: activation-aware rank allocation on top of the repaired route, backbone scale (cloud), and a collapse-boundary detector to replace the linear predictor — all against the same pre-registered gate, with certificates re-derived from the final deployed weights at every step (a post-hoc certification cost of one SVD per layer, not a redesign) and multi-seed replication as the first milestone.

VIII. CONCLUSION

qicert compresses a VLA into a form that carries an exact certificate, enforces the certificate at runtime with a guard that refuses, and measures the statistical tail honestly. The classical baseline wins the accuracy contest at its $2\times$ ceiling and we say so; above that ceiling, the training-free tensor route collapses and we say that too — with the failure mechanism isolated and used: repair training on the compressed model recovers full accuracy at $2.46\times$ under the same pre-registered gate, with certificates re-derived from the final deployed weights and the guard still enforcing. The certificate property is real, measured on every compressed layer, and structurally unavailable to the quantized baseline.

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- [3] E. Frantar et al., “GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers,” *ICLR*, 2023.

- [4] D. Grinko, J. Gacon, C. Zoufal, S. Woerner, “Iterative Quantum Amplitude Estimation,” *npj Quantum Information*, 7:52, 2021.
- [5] M. C. Campi, S. Garatti, “The Exact Feasibility of Randomized Solutions of Uncertain Convex Programs,” *SIAM J. Optimization*, 19(3):1211–1230, 2008.
- [6] J. Liu et al., “Towards Provably Efficient Quantum Algorithms for Large-Scale Machine-Learning Models,” *Nature Communications*, 15:434, 2024.

APPENDIX

Every quantitative claim above resolves to files in the public repository. Reviewers can regenerate any row from its directory.

TABLE II
CLAIM PROVENANCE.

Claim	Artifact
Reference accuracy 0.4468; split definition	results/N1v2-ckpt/baseline_seed0.json, results/eval_split.json
Training curve (1,200 steps, seed 0)	results/N1v2/*/metrics.jsonl
Calibrated INT8 0.4466 at 1.997 $\times$, CI	results/N8C/*/run.json
Uniform TT 0.000 at 2 $\times$	results/ledger.csv (N2 rows, 2026-09-12)
Repaired TT (closed-form) 0.164 at 2.54/2.86 $\times$, sound 168/168	results/N2R2/*/run.json
N9 repair training: prefix 0.1586 $\rightarrow$ 0.4519; confirm GO 0.4535 at 2.46 $\times$, CI, certs re-derived	results/N9/*/run.json (prefix), results/N9C/*/run.json (confirm), results/N9-repaired/
Layer spectra, PR 40–845	results/layer_spectra.csv
Compiler $r = 0.975$, identity 2.3×10^{-14}	results/E-comp/*/run.json, bench/compiler.py output
Latency rows (CPU profile)	results/E6-latency/*/run.json
Estimator race (p^* , widths, budgets)	results/safety/threearm_seed0.json
Conformal coverage table	results/safety/conformal_seed0.json
Guard demo (SERVE/REFUSE transcript)	results/guard-demo/transcript.json
Shadow-syndrome monitor (FPR, detection, 0.35 ms/step)	results/N10MON/*/run.json, bench/monitor.py
Output-diversity gate (collapse signature tests)	certify/guard.py (ActionDiversityMonitor), test_certify_guard.py
Lean proofs (composition + guard, zero sorry)	lean/QicertProofs/Qicert.lean
INT8 calibration ablation	results/E7-int8-ablation/*/run.json
Pre-registered gate text	N2R-SEARCH-CONFIG.md; scripts/run_remaining_experiments.py

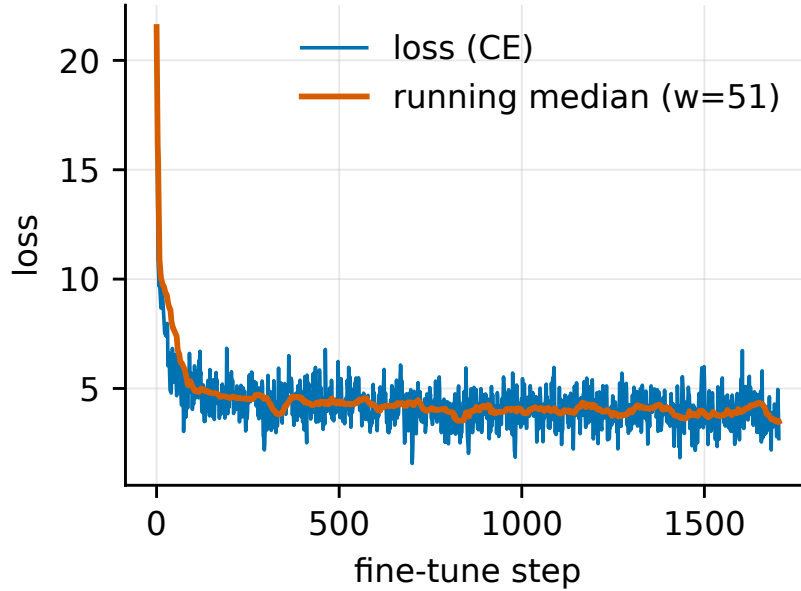


Fig. 4. Fine-tuning curve for the reference policy (seed 0): cross-entropy loss per step with running median. Loss falls from 22.5 to a median of 2.4–2.8 by step 1,200.

The capability gate was committed to the repository before the repaired compression runs: *GO* requires a full-split point with ratio $\geq 2.0\times$ and accuracy ≥ 0.3968 (reference -0.05). The verdict is computed mechanically from `run.json` artifacts by `n2r2_verdict()` in the runner; no manual adjustment is applied. The safety gates are of a different kind and were checked rather than gated: certificate soundness on every compressed layer (168/168 across all repaired runs), guard refusal on all four tampering scenarios, and conformal coverage inside the finite-sample band at all tested levels.

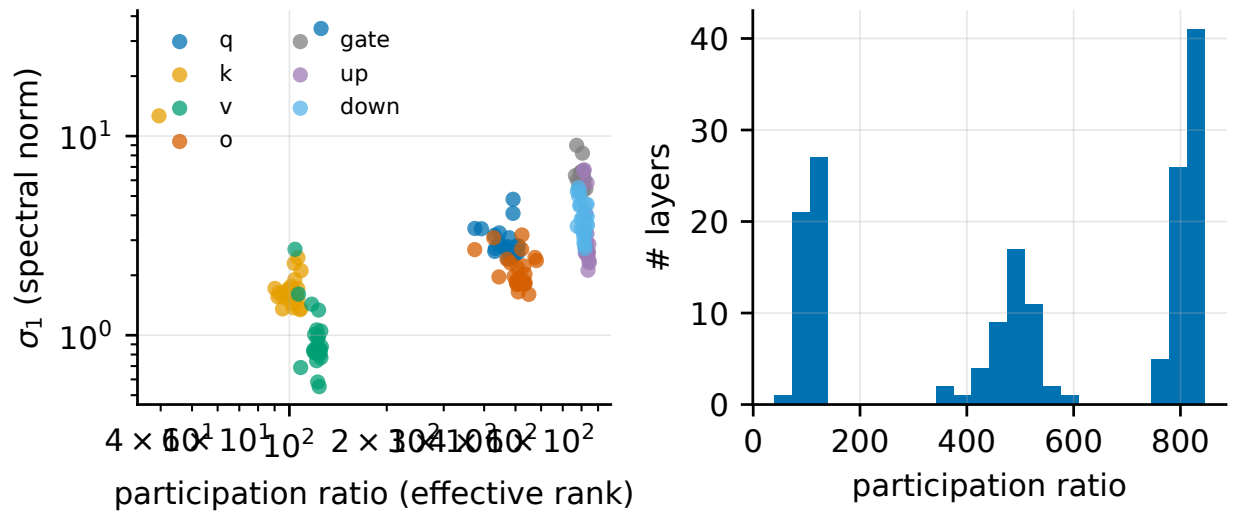


Fig. 5. Spectral profile of the 168 linear layers the certificates are computed from (left: participation ratio vs. spectral norm, per projection type; right: participation-ratio histogram). High effective ranks explain why uniform truncation removes signal rather than redundancy.

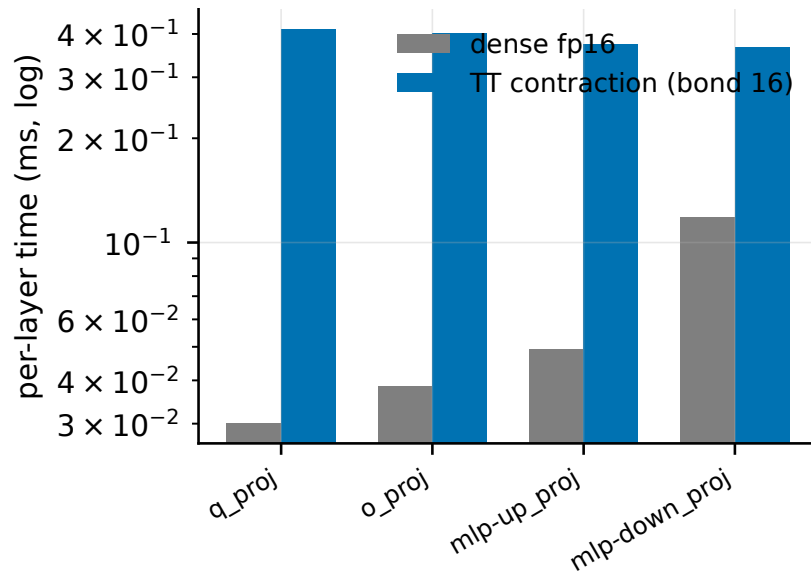


Fig. 6. Measured per-layer contraction time on the CPU edge profile (bond 16): dense fp16 vs. TT. All measured paths stay within the 100 ms budget; the slowdown factor is reported rather than hidden.

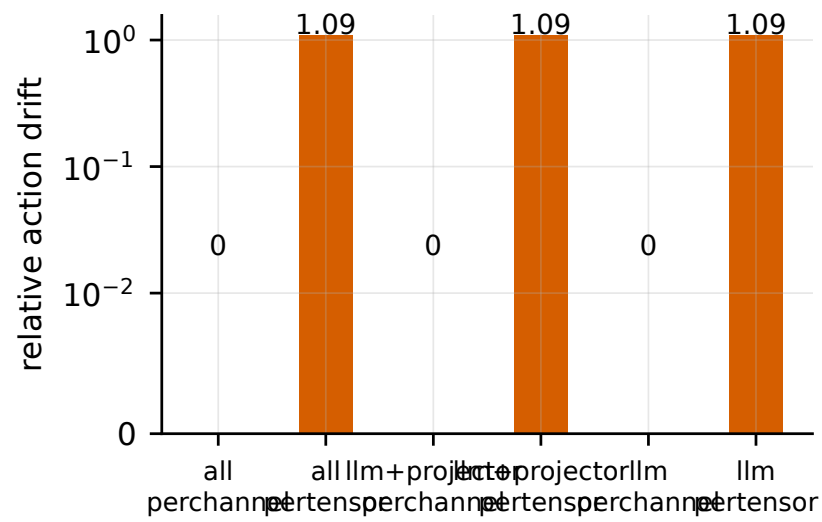


Fig. 7. Calibration ablation for the INT8 baseline: per-channel MSE calibration holds relative action drift at zero where per-tensor scaling drifts by 1.09. The naive uncalibrated 0.000 INT8 figure quoted elsewhere is this artifact, not a property of INT8.